

**Experiential Learning
Project Assessment-I
on
“Rainfall Data Analyzer”
under the course
Python Programming Language (N-PCCCM304P)**

Presented By

Swanand S. Barve



**Under the Guidance
of**

Mrs. Mayuri Getme

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Department of CSE(AI&ML)

**S. B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT AND
RESEARCH,NAGPUR**

An Autonomous Institute, Affiliated to RTMNU, Nagpur

Problem Statement

Develop a Python application to record and analyze rainfall data for different cities or months. The system should calculate total rainfall, average rainfall, maximum and minimum rainfall, and generate summary reports and graphical visualizations.

❖ Total Rainfall

Sum of all recorded rainfall values across the dataset

❖ Average Rainfall

Mean value to understand typical rainfall patterns

❖ Maximum Rainfall

Identify the highest recorded rainfall event

❖ Minimum Rainfall

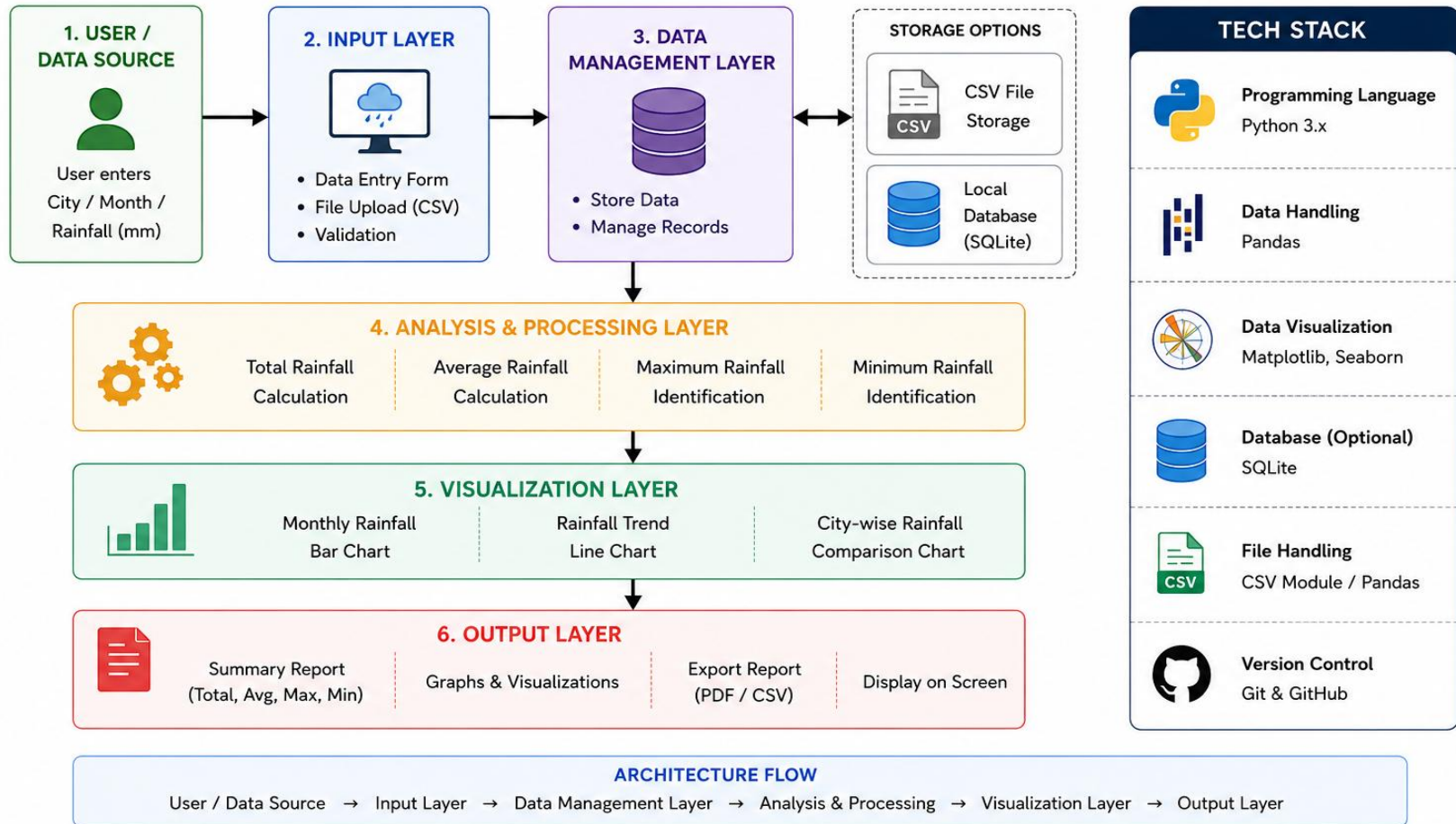
Identify the lowest recorded rainfall event

The system should also generate a summary report and graphical representation for easy interpretation

System Architecture & TechStack

RAINFALL DATA ANALYZER

SYSTEM ARCHITECTURE WITH TECH STACK



Objectives



Objectives

The main objectives of the Rainfall Data Analyzer are:

1

Record Rainfall Data

Accept rainfall data for different cities or months from the user.

2

Store in Structured Format

Organize collected data using Python lists and data structures.

3

Calculate Statistics

Compute total, average, maximum, and minimum rainfall values.

4

Report & Visualize

Generate a summary report and graphs for easy understanding.

Algorithm & Pseudocode



01 Start the program

02 Enter Data

Input city/month and corresponding rainfall values.

03 Store Data

Save rainfall values in a list or suitable data structure

04 Analyze Data

Calculate total, average, maximum, and minimum rainfall

05 Display Results

Show all calculated statistics to the user

06 Generate Graph & Report

Create graphical visualization and summary report, report, then stop

Pseudocode

START

Input rainfall values

```
total = sum(rainfall)
```

```
average = total / number of values
```

```
maximum = max(rainfall)
```

```
minimum = min(rainfall)
```

Display total, average,
maximum, minimum

Generate rainfall graph

END

Understanding of Python Logic

Python Concept	Purpose
Variables	Store rainfall values and results
Lists	Store multiple rainfall values
Input	Accept rainfall data from user
Loops	Process multiple entries
Functions	Organize program logic
sum()	Calculate total rainfall
max() / min()	Find maximum / minimum rainfall
Matplotlib	Create graphs and visualizations



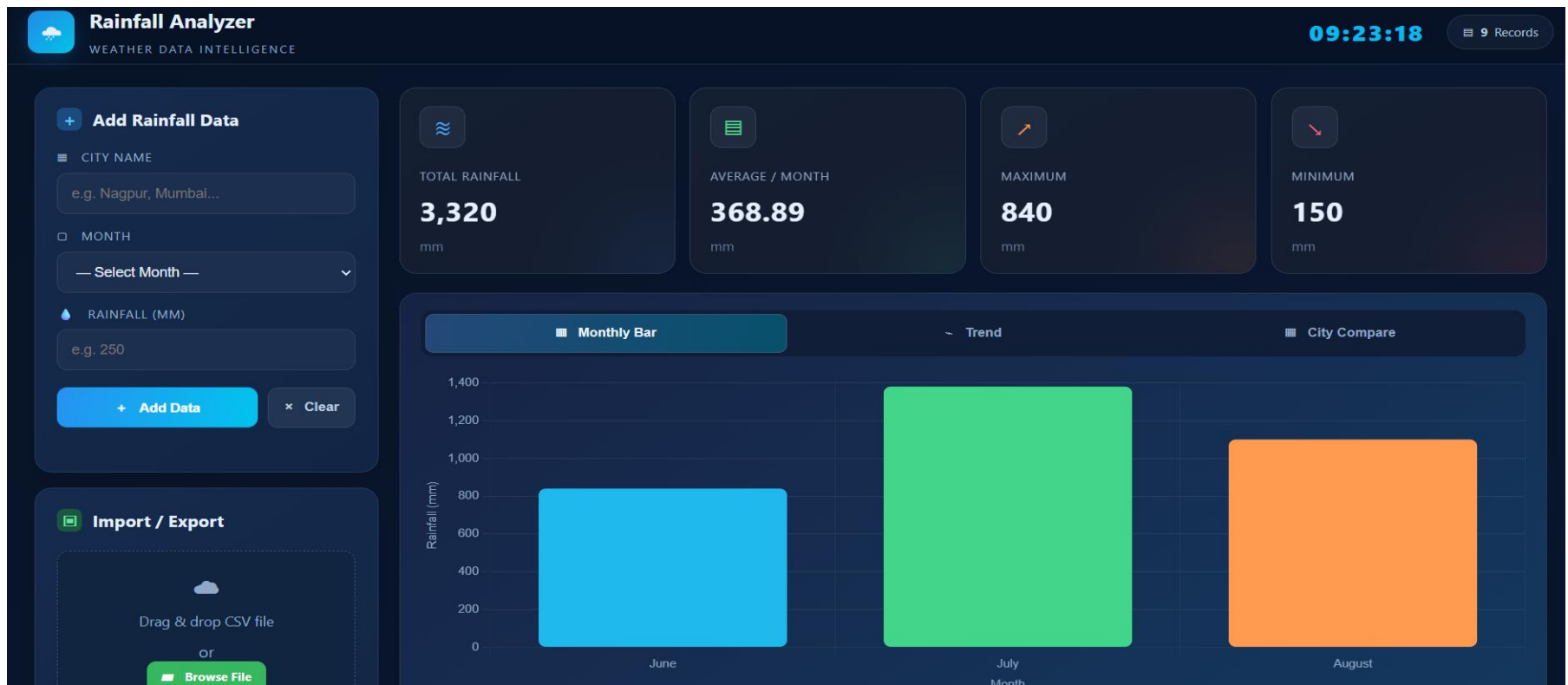
Phase 1 vs Phase 2 – Project Progress

Feature		Phase 1	Phase 2
	Problem identification	Problem identification	Working implementation
	Basic Python logic	Basic Python logic	Interactive dashboard
	Pseudocode	Pseudocode	Actual data processing
	Basic graph	Basic graph	Multiple interactive charts
	Manual rainfall input	Manual rainfall input	Form + CSV import
	Basic calculations	Basic calculations	Automatic statistics
	Python/Matplotlib	Python/Matplotlib	HTML + CSS + JavaScript + Chart.js



GitHub Account – Swanand-GH

Repository - [rainfall-data-analyzer\(Python\)](#)



Thank You